



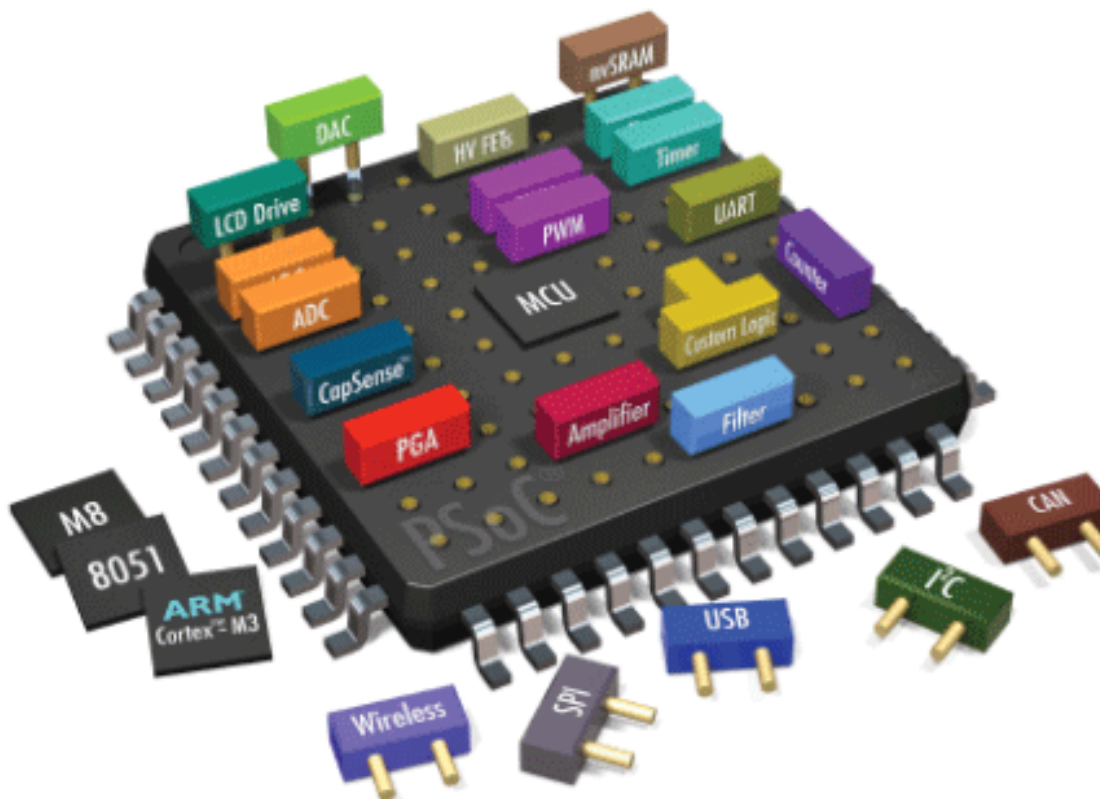
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DEPARTMENT OF COMPUTER ENGINEERING

Lab # 08: Demonstration and practice of 7-Segment Display Control and display different patterns.

Embedded Systems

Demonstration and practice of 7-Segment Display Control and display different patterns.



Dated:

Week 08

Semester:

Fall 2024



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8051 microcontroller:

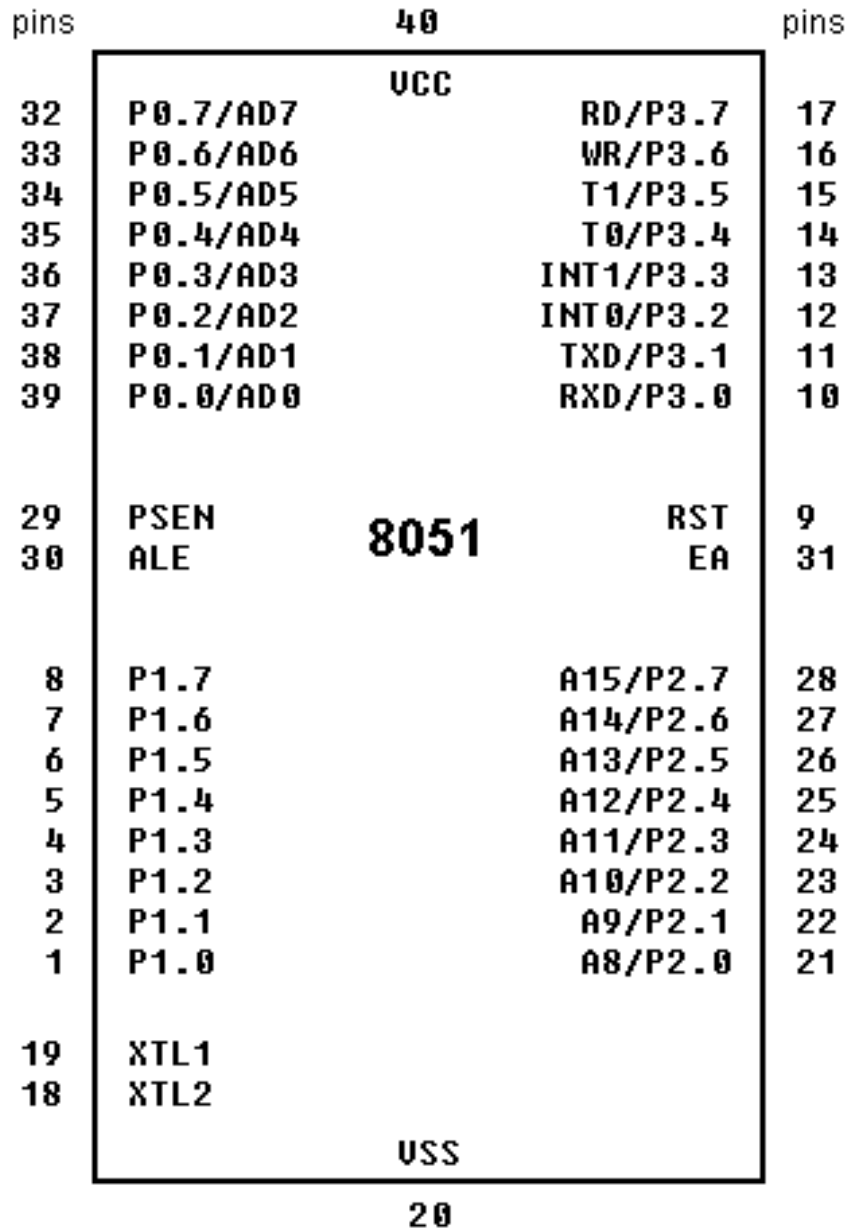
8051 microcontroller is designed by Intel in 1981. It is an 8-bit microcontroller. It is built with 40 pins DIP (dual inline package), 4kb of ROM storage and 128 bytes of RAM storage, 2 16-bit timers. It consists of are four parallel 8-bit ports, which are programmable as well as addressable as per the requirement. An on-chip crystal oscillator is integrated in the microcontroller having crystal frequency of 12 MHz.



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- **Pins 1 to 8** – These pins are known as Port 1. This port doesn't serve any other functions. It is internally pulled up, bi-directional I/O port.
- **Pin 9** – It is a RESET pin, which is used to reset the microcontroller to its initial values.
- **Pins 10 to 17** – These pins are known as Port 3. This port serves some functions like interrupts, timer input, control signals, serial communication signals RxD and TxD, etc.



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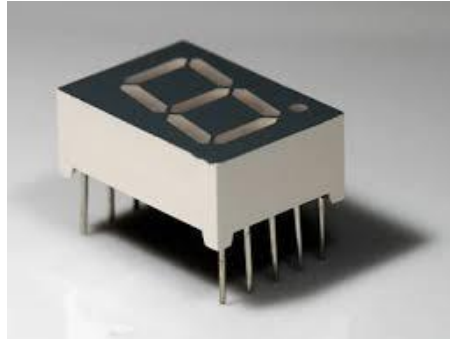
Pin	Name	Bit Address	Function
P3.0	RXD	B0H	Receive data for serial port
P3.1	TXD	B1H	Transmit data for serial port
P3.2	INT0-bar	B2H	External interrupt 0
P3.3	INT1-bar	B3H	External interrupt 1
P3.4	T0	B4H	Timer/counter 0 external input
P3.5	T1	B5H	Timer/counter 1 external input
P3.6	WR-bar	B6H	External data memory write strobe
P3.7	RD-bar	B7H	External data memory read strobe

- **Pins 18 & 19** – These pins are used for interfacing an external crystal to get the system clock.
- **Pin 20** – This pin provides the power supply to the circuit.
- **Pins 21 to 28** – These pins are known as Port 2. It serves as I/O port. Higher order address bus signals are also multiplexed using this port.
- **Pin 29** – This is PSEN pin which stands for Program Store Enable. It is used to read a signal from the external program memory.
- **Pin 30** – This is EA pin which stands for External Access input. It is used to enable/disable the external memory interfacing.
- **Pin 31** – This is ALE pin which stands for Address Latch Enable. It is used to demultiplex the address-data signal of port.
- **Pins 32 to 39** – These pins are known as Port 0. It serves as I/O port. Lower order address and data bus signals are multiplexed using this port.
- **Pin 40** – This pin is used to provide power supply to the circuit.

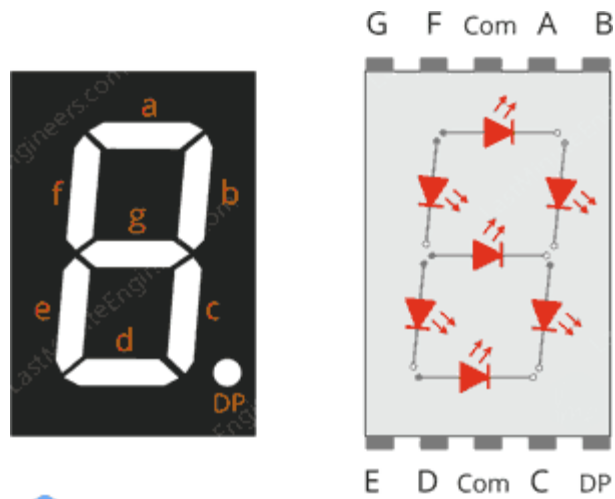
The 7-segment Displays:



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The 7-segment displays are really just seven LEDs lined up in a particular pattern. In this case, the number '8' shape we're all familiar with. Each of the seven LEDs is called a segment because when illuminated the segment forms part of a numerical digit (both Decimal and Hex) to be displayed. An additional 8th LED is sometimes used for indication of a decimal point.



Each one of the seven LEDs in the display is given a positional segment with one of its connection pins being brought straight out of the rectangular plastic package. These individual LED pins are labeled from a through to g representing each individual LED. The other LED pins are connected together and wired to form a common pin.

To turn on and off a particular part of the display, you set the appropriate pin HIGH or LOW just like you would with a regular LED. So that some segments will be light and others will be dark allowing the desired character pattern of the number to be generated on

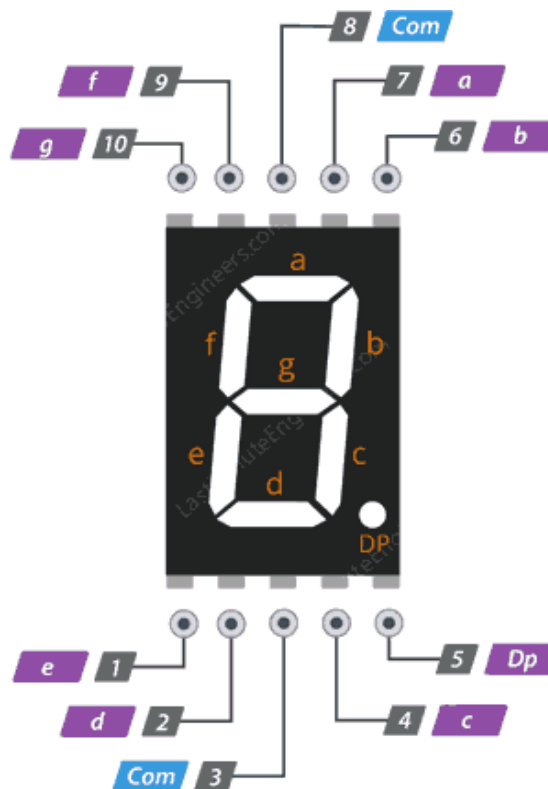


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the display. This then allows us to display each of the ten decimal digits 0 through to 9 on the same 7-segment display.

7 Segment Display Pinout

Now let's go over the segment configuration so we know which pins light up which segments. The pinout for the 7-segment display is as follows.



a-g & DP Out of 10, the 8 pins i.e. a, b, c, d, e, f, g and DP segment (decimal point) are connected to digital pins of Arduino. By controlling each LED on the segment connected, numbers can be displayed.

COM The pin 3 and 8 are internally connected to form a common pin. This pin should be connected to GND (common cathode) or 5V (common anode) depending upon the type of the display.

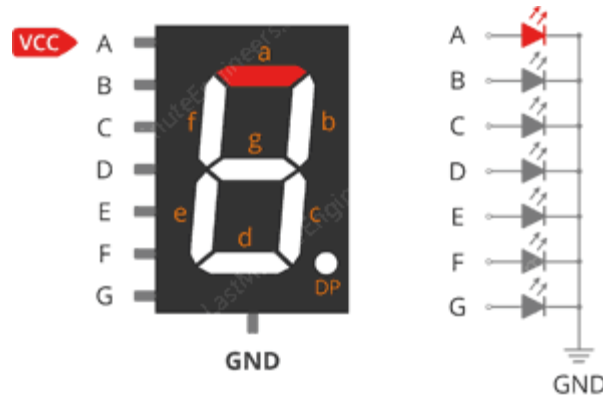


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Common Cathode(CC) Vs Common Anode(CA)

Seven segment displays are of two types: Common Cathode (CC) and Common Anode (CA). The Internal structure of both types is nearly the same. The difference is the polarity of the LEDs and common terminal. As their name suggests, the common cathode has all the cathodes of the LEDs in a 7-segment connected together and the common anode has all the anodes of the LEDs in a 7-segment connected together.

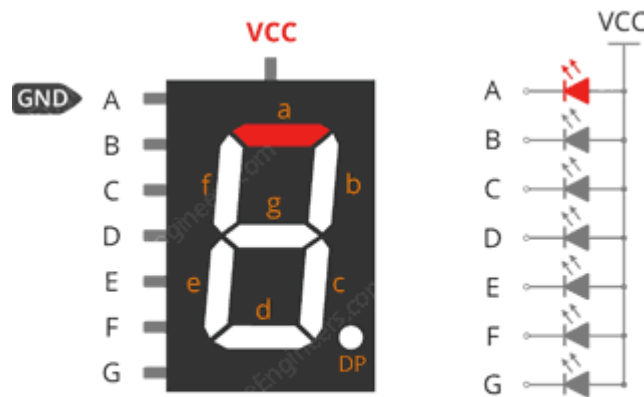
In the common cathode display, all the cathode connections of the LED segments are connected together to 'logic 0' / GND. The individual segments are then illuminated by applying HIGH / 'logic 1' signal to the individual Anode terminals (a-g).



Common Cathode 7 Segment Working

In the common anode display, all the anode connections of the LED segments are joined together to logic "1". The individual segments are illuminated by applying a ground, logic "0" or "LOW" signal to the Cathode of the particular segment (a-g).

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Common Anode 7 Segment Working

How 7 Segment Display Works?

Depending upon the decimal digit to be displayed, the particular set of LEDs is illuminated. For instance, to display the numerical digit 4, we will need to light up four of the LED segments corresponding to b, c, f and g. Thus the various digits from '0 through 9' and characters from 'A through F' can be displayed using a 7-segment display as shown.



Below truth table shows the individual segments that need to be illuminated in order to produce digits and characters. Please note that truth table for common anode 7-segment display is exact opposite to that of common cathode 7-segment display.



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Common Cathode Seven Segment Truth-Table

	a	b	c	d	e	f	g
0	On	On	On	On	On	On	Off
1	Off	On	On	Off	Off	Off	Off
2	On	On	Off	On	On	Off	On
3	On	On	On	On	Off	Off	On
4	Off	On	On	Off	Off	On	On
5	On	Off	On	On	Off	On	On
6	On	Off	On	On	On	On	On
7	On	On	On	Off	Off	Off	Off
8	On	On	On	On	On	On	On
9	On	On	On	Off	Off	On	On
A	On	On	On	Off	On	On	On
B	Off	Off	On	On	On	On	On
C	On	Off	Off	On	On	On	Off
D	Off	On	On	On	On	Off	On
E	On	Off	Off	On	On	On	On
F	On	Off	Off	Off	On	On	On



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Common Anode Seven Segment Truth-Table

	a	b	c	d	e	f	g
0	Off	Off	Off	Off	Off	Off	On
1	On	Off	Off	On	On	On	On
2	Off	Off	On	Off	Off	On	Off
3	Off	Off	Off	Off	On	On	Off
4	On	Off	Off	On	On	Off	Off
5	Off	On	Off	Off	On	Off	Off
6	Off	On	Off	Off	Off	Off	Off
7	Off	Off	Off	On	On	On	On
8	Off	Off	Off	Off	Off	Off	Off
9	Off	Off	Off	On	On	Off	Off
A	Off	Off	Off	On	Off	Off	Off
B	On	On	Off	Off	Off	Off	Off
C	Off	On	On	Off	Off	Off	On
D	On	Off	Off	Off	Off	On	Off
E	Off	On	On	Off	Off	Off	Off
F	Off	On	On	On	Off	Off	Off

74LS47 BCD to 7-Segment Display Decoder IC:

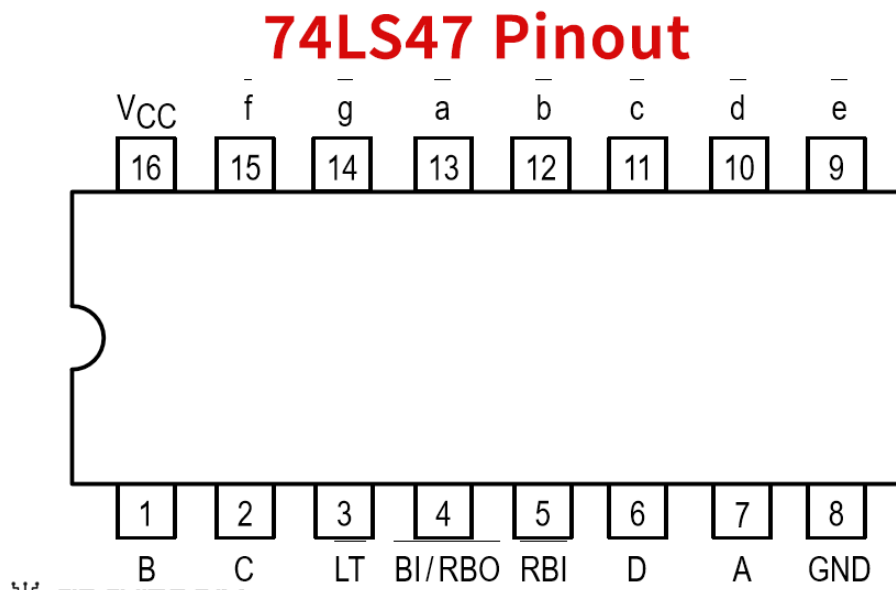


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A BCD to Seven Segment decoder is a combinational logic circuit that accepts a decimal digit in BCD (input) and generates appropriate outputs for the segments to display the input decimal digit.

74LS47 Pinout



74LS47 Pin Configuration

Pin No	Pin Name	Description
1	B	BCD input of the IC
2	C	BCD input of the IC
3	Display test/Lamp test	To test the display LED
4	Blank Input	To turn-off the LEDs of the display



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5	Store	Store or strobe a BCD code
6	D	BCD input of the IC
7	A	BCD input of the IC
8	GND	Ground Pin
9	e	7 segment output 1
10	d	7 segment output 2
11	c	7 segment output 3
12	b	7 segment output 4
13	a	7 segment output 5
14	g	7 segment output 6
15	f	7 segment output 7
16	VCC	Supply Voltage

Binary Coded Decimal

Binary Coded Decimal (BCD or “8421” BCD) numbers are made up using just 4 data bits (a nibble or half a byte) similar to the Hexadecimal numbers we saw in the binary tutorial, but unlike hexadecimal numbers that range in full from 0 through to F, BCD numbers only range from 0 to 9, with the binary number patterns of 1010 through to 1111 (A to F) being invalid inputs for this type of display and so are not used as shown below.

Decimal	Binary Pattern				BCD
	8	4	2	1	
0	0	0	0	0	0
1	0	0	0	1	1
2	0	0	1	0	2
3	0	0	1	1	3
4	0	1	0	0	4
5	0	1	0	1	5
6	0	1	1	0	6
7	0	1	1	1	7
8	1	0	0	0	8
9	1	0	0	1	9
10	1	0	1	0	Invalid
11	1	0	1	1	Invalid
12	1	1	0	0	Invalid
13	1	1	0	1	Invalid
14	1	1	1	0	Invalid
15	1	1	1	1	Invalid



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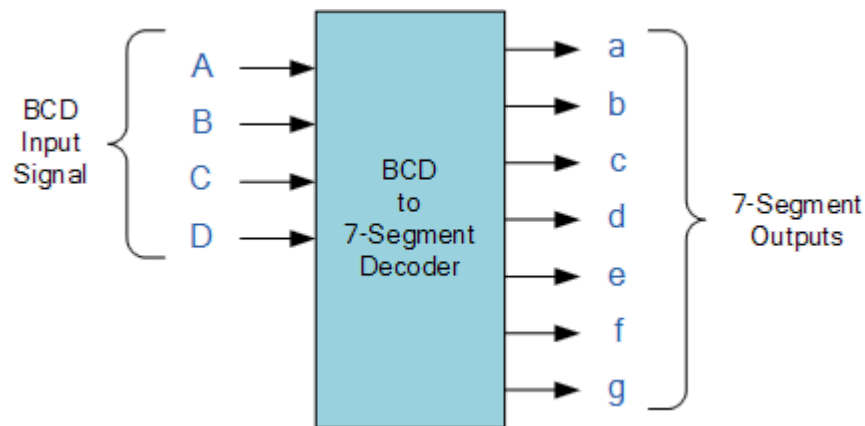
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BCD to 7-Segment Display Decoders

A binary coded decimal (BCD) to 7-segment display decoder such as the TTL 74LS47 or 74LS48, have 4 BCD inputs and 7 output lines, one for each LED segment. This allows a smaller 4-bit binary number (half a byte) to be used to display all the denary numbers from 0 to 9 and by adding two displays together, a full range of numbers from 00 to 99 can be displayed with just a single byte of eight data bits.

BCD to 7-Segment Decoder



The use of **packed** BCD allows two BCD digits to be stored within a single byte (8-bits) of data, allowing a single data byte to hold a BCD number in the range of 00 to 99.

Example:

A program which act as up counter from 00 to 99 using the two 7-segment displays.

```
ORG 000H
```

```
MOV A,#00H
```



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NEXT:

```
MOV P2,A
CALL DELAY
ADD A,#1
DA A
```

JMP NEXT

```
;=====
; DELAY 0.5S
;=====
```

DELAY:

```
MOV R5,#5
DL2:
    MOV R6,#200
    DL1:
        MOV R7,#249
        DJNZ R7,$
        DJNZ R6,DL1
    DJNZ R5,DL2
```

RET

END

Lab Tasks:

- 1. Write a program which act as up and down counter from 0 to 9 and 9 to 0 using any one of the 7-segment display.**
- 2. Write a program which act as up and down counter from 00 to 99 and 99 to 00 using the two 7-segment displays.**
- 3. Write a program which sequentially and repeatedly show the numbers 90 through 09 on DS3 by decrement and on DS4 by increment. Each number stays on 0.5 seconds.**
- 4. Write a program which make your 7-degment display to act as a Dice upon pressing a switch.**

Hint:

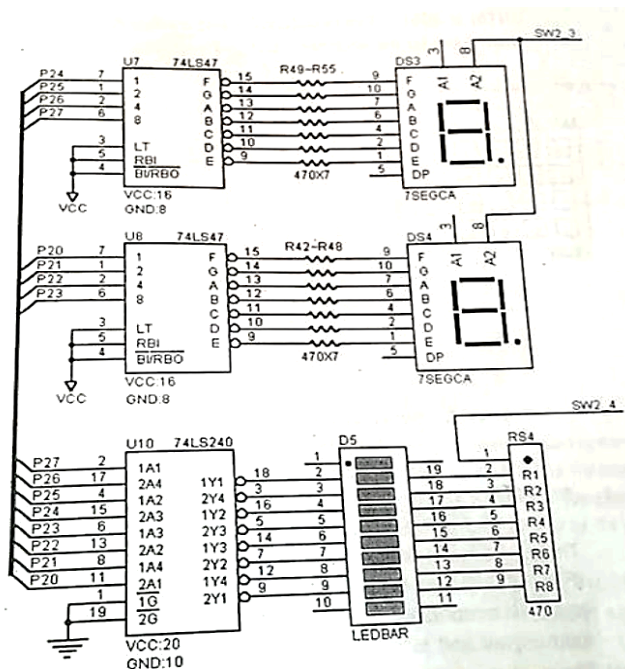
1. The two-digit 7-segment display (DS3 – DS4) of MTS-51 Trainer to show decimal numbers. Since the BCD-to-7-segment decoder/drivers are built in the Trainer,



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the BCD value to be displayed should be written to port 2. The I/O lines of P2.3 - P2.0 are the inputs of DS4 (units' digit) and P2.7 - P2.4 lines are the inputs of DS3 (tens digit). Note that the SW2_3 switch should be placed in ON position.

2. Circuit diagram



Lab Performance Evaluation:

Read and practice the manual. Performance is evaluated at the end of lab based on successfully running and understanding of examples, tasks and viva using defined rubrics.

Lab Report Evaluation:

Submit the Lab Report by writing all the examples and tasks which must include the following:

1. Brief description about how its work (3-5 lines)
2. The code
3. The results (Screenshot)

Important note: Lab Report must be according to the Lab Report Format available on Google Classroom and must be submitted on time. Two similar reports will get zero marks. So, don't try to copy your fellow reports. Lab report must be submitted in group of three maximum.